



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

Name : Atharva Ambilwade

PRN : 25070122055

Class : CSE

Batch : 2025-29

Assignment :

Subject: Programming with Java

ASSIGNMENT No : 9

TITLE : Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Problem Statement

Write a Java program to demonstrate the use of the final keyword with variables, methods, and classes.

Learning Objectives

To implement the final keyword with variables, methods, and classes.

Theory

The final keyword restricts modification in Java programs. A final variable cannot be reassigned after initialization. A final method cannot be overridden by subclasses, while a final class cannot be inherited. This improves security, prevents accidental modifications, and increases program reliability. Final is commonly used for constants and immutable classes.



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Code and Output Screenshots

```
J Main10.java > Java > Main10
1  final class College {
2      void collegeName() {
3          System.out.println("Current College : Symbiosis Institute of Technology");
4      }
5  }
6  class Person {
7      final int id = 55;
8      final void displayId() {
9          System.out.println("ID : " + id);
10     }
11 }
12 class Student extends Person {
13     String name;
14     Student(String name) {
15         this.name = name;
16     }
17     void displayStudent() {
18         System.out.println("Name : " + name);
19     }
20 }
21 public class Main10 {
22     Run main | Debug main | Run | Debug
23     public static void main(String[] args) {
24         Student s = new Student("Atharva");
25         s.displayStudent();
26         s.displayId();
27         College c = new College();
28         c.collegeName();
29     }
}
```

PROBLEMS 14

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Filter (e.g. text, lexcludeText, t...

[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Main10.java && java Main10

Name : Atharva

ID : 55

Current College : Symbiosis Institute of Technology

[Done] exited with code=0 in 0.542 seconds



Conclusion

This program demonstrates the use of the final keyword in Java. It shows that a final class cannot be inherited, a final variable cannot be modified, and a final method cannot be overridden. Thus, the final keyword helps ensure data security and prevents unintended changes in a Java program.

Exercise

1. Create a Bank Account program where account number is final and cannot be changed.

```
J Main.java > Java > Main
1  class BankAccount {
2      final int accountNumber = 12345;
3      String name;
4      BankAccount(String name) {
5          this.name = name;
6      }
7      void display() {
8          System.out.println("Name: " + name);
9          System.out.println("Account Number: " + accountNumber);
10     }
11 }
12 public class Main {
13     Run main | Debug main | Run | Debug
14     public static void main(String[] args) {
15         BankAccount b = new BankAccount("Atharva");
16         b.display();
17     }
18 }
```

PROBLEMS 14 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Filter (e.g. te

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Main.java && java Main
Name: Atharva
```

```
Account Number: 12345
```

```
[Done] exited with code=0 in 0.548 seconds
```



2. Create a Library Book Management program where the book ISBN is declared as final and cannot be changed once assigned. Display the book's ISBN, title, author, and price.

```
J Main8.java > Java > Main8
1  class Book {
2      final String isbn = "978-1234567890";
3      String title;
4      String author;
5      double price;
6      Book(String title, String author, double price) {
7          this.title = title;
8          this.author = author;
9          this.price = price;
10     }
11     void display() {
12         System.out.println("ISBN: " + isbn);
13         System.out.println("Title: " + title);
14         System.out.println("Author: " + author);
15         System.out.println("Price: Rs. " + price);
16     }
17 }
18 public class Main8 {
19     Run main | Debug main | Run | Debug
20     public static void main(String[] args) {
21         Book b = new Book("The Alchemist", "Paulo Coelho", 699);
22         b.display();
23     }
24 }
```

PROBLEMS 14 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Filter (e.g. text, !excludeText, t...

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Main8.java && java Main8
ISBN: 978-1234567890
Title: The Alchemist
Author: Paulo Coelho
Price: Rs. 699.0
|
[Done] exited with code=0 in 0.493 seconds
```



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